

RV32I CPU

Verification Specification

Reproducible stimuli, checks and waveform evidence

Document	RV32I CPU — Verification Specification
Document type	Verification specification
Subject	Reproducible stimuli, checks and waveform evidence
Revision	2.2
Date	23 September 2026
Author	Bunea Cosmin-Andrei
Status	Released for review

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1 Introduction

1.1 Purpose of the document

This specification describes the simulation strategy, stimulus, reference models and acceptance checks of the RV32I three-stage CPU. It is intended for hardware designers, verification engineers and software developers integrating the block.

It defines the behaviour that the RTL must show at its interfaces.

1.2 Validity of the document

The documented configuration is the RTL in `cpu/hdl` and the associated CPU/PIC and SoC simulation environments. Interface adaptations from the supplied briefs are identified in the integration section.

Revision 2.2 of 23 September 2026 describes the delivered state of that RTL.

1.3 Definitions of terms and abbreviations

Term	Meaning
AXI4-Lite	Single-beat AXI read/write channels.
BHT / BTB	Branch direction / target tables.
CSR	Control and status register.
DUT	Device under test.
EOI	End of interrupt; completes one service level.
IRQ	Interrupt request.
LSU	Load/store unit.
NBA	Nonblocking-assignment update region in simulation.
PIC	Programmable interrupt controller.
RAS	Return-address stack.
S1 / S2 / S3	Fetch / decode-execute / writeback.
SVA	SystemVerilog Assertions.

1.4 Relationship with other documents

The documents below are the ones this specification is derived from or read with.

Document	Version / date	Author	Relationship
RV32I 3-Stage Pipeline CPU	Draft, 1 July 2026	D. Borzasi, Siemens	Functional requirements; see section 2.
Spec table of contents and templates	2 September 2026	D. Borzasi, Siemens	Document structure and verification-plan template.
Design Specification — CPU	Rev. 2.2, 23 September 2026	C.-A. Bunea	Companion document.

Document	Version / date	Author	Relationship
RISC-V Unprivileged ISA	20191213	RISC-V International	Instruction semantics.
RISC-V Privileged Architecture	20211203	RISC-V International	Machine-mode CSRs and traps.
AMBA AXI and ACE Protocol Specification	IHI0022E	Arm	AXI4-Lite ports.
IEEE Std 1364 (Verilog)	2005	IEEE	RTL language.

1.5 Notes on the notation used

Signal and register names follow the RTL. Addresses and waveform bus values are hexadecimal; time is in ns and widths are in bits. Register offsets are byte offsets; CSR addresses are instruction-encoded indices.

Notation	Meaning
Fixed-width name	A Verilog module, port, file or script name, spelled as in the source.
Signal suffix <code>_i</code> / <code>_o</code>	Module input / output, seen from inside that module.
<code>[a:b]</code>	Bit range, most significant bit first.
<code>S1</code> / <code>S2</code> / <code>S3</code>	Pipeline stage names used throughout the document.
<code>posedge +1 ns</code>	A testbench stimulus time, one nanosecond after the sampling edge.

2 Verification methodology

2.1 Overview

A run either passes or fails; this section states what that verdict covers. The first table gives the acceptance conditions applied to every test and the limits of what simulation shows; the second gives the runs behind the results quoted in this document.

Item	Contract
Requirement source	RV32I 3-Stage Pipeline CPU.pdf
Device under test	rv32i_cpu_top and its RTL modules.
Functional verdict	Every expected value and response must match; errors=0 and test_done=1 are required.
Protocol verdict	VALID/payload stability, response ordering and outstanding transaction limits; clocked at posedge.
Failure handling	Mismatch, timeout, incomplete test, assertion failure or coverage-gate failure returns a failing run.
Limits of evidence	Simulation against independent reference models and the official ISA tests; no synthesis or timing result.

Simulator / flow	Recorded evidence (23 September 2026)
ModelSim	39 CPU/PIC runs; zero mismatches.
Verilator -assert	17 CPU/PIC benches in 53 runs; merged functional coverage 90/90 bins and every cover property reached.
Integrated SoC	34 ModelSim runs; 23 assertion-enabled benches in 51 Verilator runs.
riscv-tests	53 of 58 rv32ui/rv32mi tests pass; the other five need features outside the implemented ISA.
Mutation check	25 injected RTL defects, each detected by its bench.
PULP comparison	Decoder over ten random seeds plus 96 register-slave strobe/order/stall cases and RO/unmapped/reset cases.

2.2 Implementation strategy

Directed tests exercise individual features and simultaneous events; program and reference tests check end-to-end results. Protocol checks observe accepted transfers; state checks sample after nonblocking updates.

Seeded random programs and the official riscv-tests extend the directed set. Every bench also runs under Verilator with the assertions bound.

2.3 Verification plans

The plan in section 4.1 follows the supplied template: ID, title, description, status and test/checker, one row per test.

2.4 Reference model

`isa_reference.py` encodes and interprets RV32I in Python, independently of the RTL and the assembler. For each program it writes the image, the expected PC, destination and value of every completed instruction, and the final data memory. The official riscv-tests are an external reference; each test reports its result through tohost.

3 Verification environment

3.1 Overview

The program image supplies instructions and initial data to the responders. Checks compare observed architectural state with expected results, while protocol monitors and assertions inspect the bus and pipeline independently.

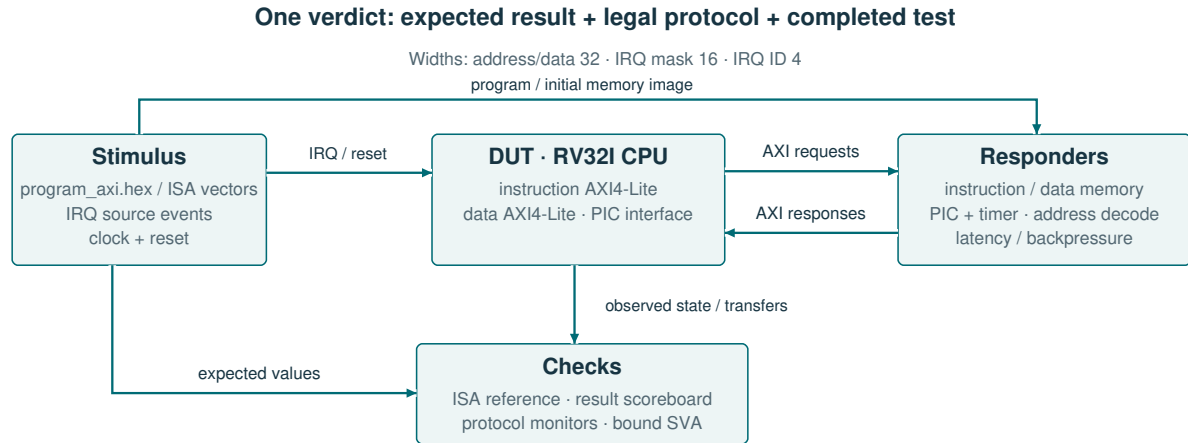


Figure 1: Test stimuli, DUT and independent checking paths.

Component	File / role
Clock / reset	ck_rst_tb.v: shared generator; dedicated reset benches can drive a reset pulse between edges.
Bus stimulus	tb_axil_master.vh: register-access tasks; CPU programs supply real load/store requests at system level.
Memory model	axi_lite_mem_model.v: configurable response delay and repeatable backpressure.
Reference	isa_reference.py: instruction trace.
Verdict	tb_check.vh: explicit mismatch count, completion and fatal failure.
Protocol	axi_lite_monitor.v for ModelSim; axi_lite_sva.sv under Verilator.

3.2 Control and drivers

Stimulus is driven one nanosecond after the sampling edge, so that a test never races the design it is checking. The table gives the timing rule for each driven event and the reason for it; the waveform shows the reset case, the one event placed off the clock grid.

Signal / event	Timing rule	Reason
clk	10 ns period in these traces; rising edges are 5, 15, 25... ns.	Defines the handshake observation points.
Shared bus / claim tasks	Drive 1 ns after posedge, in a separate statement. Hold until the sampling edge.	Avoid blocking-assignment races with DUT active/NBA evaluation.

Signal / event	Timing rule	Reason
Handshake observation	Sample VALID && READY at posedge before changing stimulus.	Records the transfer that the DUT accepted.
Reset assertion	Dedicated reset tests assert at posedge +2 ns, check at +3 ns.	The check occurs before either the next falling or rising edge.
Reset release	Explicit nanosecond delays, off the clock grid; shared generator releases at 123 ns.	Exercises asynchronous stimulus without a coincident clock event.

Reset is asserted while an instruction request is waiting. ARVALID clears at the reset event, before the next rising edge; after release the fetch request resumes from RESET_PC.

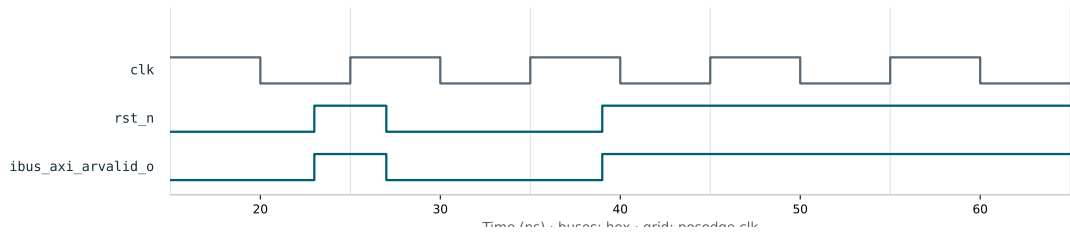


Figure 2: Reset from active state at posedge +2 ns; cleared state is checked at +3 ns.

3.2.1 Clock and reset generator

ck_rst_tb.v generates a 10 ns clock and holds reset from time 0 to 123 ns, off the clock edge. rv32i_tb_reset drives its own reset pulses between edges.

3.2.2 Memory models

axi_lite_mem_model.v serves the instruction and data ports. READ_LAT and WRITE_LAT add fixed latency; STALL_PROB and SEED add seeded random backpressure. Overrides are passed with -G and read back after elaboration.

3.2.3 Programs

program_axi.s, assembled by asm.py, runs on the system bench. isa_reference.py writes the directed ISA program and, with -random-set, ten random programs, each with its expected trace. run_riscv_tests.sh builds the official tests with a RISC-V GCC and links them at address 0.

3.3 Monitors and checkers

Checking is split between bus protocol and functional result, so that a failure points at one of the two.

3.3.1 AXI protocol monitor

axi_lite_monitor.v checks every transfer on the monitored ports in ModelSim and counts transactions; a run without traffic fails.

3.3.2 Assertions

`axi_lite_sva.sv` checks channel stability, legal responses, ordering and one outstanding transaction per direction; `rv32i_cpu_core_sva.sv` checks pipeline, trap and forwarding invariants. `rv32i_bind_core_sva.sv` attaches the checkers to the RTL module types with `bind`; they only read DUT signals. Verilator runs them with `-assert`; ModelSim ASE runs the procedural monitors.

3.3.3 Functional coverage

`rv32i_bind_sva.sv` attaches the coverage model `rv32i_cpu_func_cov.sv`. Its bin counts are merged over the system and ISA runs, and every bin must be reached; every cover property must be reached as well.

3.3.4 Result checkers

`tb_check.vh` counts mismatches and ends each run with an explicit verdict. `rv32i_tb_isa` compares the PC, destination and value of every completed instruction with the reference trace, then the whole data memory.

3.3.5 Non-vacuous checks

Non-vacuous check	Evidence
Valid traffic	Read/write counts and exercised tests prevent a disconnected monitor from being counted as coverage.
Preconditions	RO-write test first creates nonzero state; async-reset test first creates pending/nonzero state.
Negative scenarios	Watch an interval for forbidden offers/claims; do not infer absence from one sample.

3.4 Simulation environment setup

The environment is a fixed folder layout plus a small number of named scripts. The tables give that layout, the exact command for each step in the order it is run, and the tools each step needs.

3.4.1 Structure

Folder	Contents
<code>cpu/hdl</code>	Synthesizable RTL: the CPU modules, the PIC, the timer and the AXI4-Lite register front-end.
<code>cpu/debug/hdl</code>	Testbenches, bus masters and the configurable memory and responder models.
<code>cpu/debug/sva</code>	Assertion modules and the bind files that attach them to RTL module types.
<code>cpu/debug/sim</code>	Run scripts, file lists, reference models, generated programs and run logs.
<code>soc/hdl</code>	Integration fabric: address decode, arbitration, AXI4-Full to Lite conversion and RAM.
<code>soc/debug/hdl, sva, sim</code>	The same split as above, applied to the integrated SoC.

Folder	Contents
cpu/docs	Specification sources, generated figures and the waveform windows.

3.4.2 Scripts

Step	Working directory	Command / script	Result
1	Repository root	make asm	Regenerate programs and independent reference vectors.
2	cpu/debug/sim	vsim -c -do "do regress.do; quit -f"	Compile and run 39 CPU/PIC runs.
3	cpu/debug/sim (Bash / WSL)	bash run_verilator.sh	Lint, 17 benches in 53 runs with SVA, coverage and cover gates.
4	soc/debug/sim	vsim -c -do "do regress.do; quit -f"	34 integration runs.
5	Repository root (Bash / WSL)	bash soc/debug/sim/run_verilator.sh	SoC lint and 23 assertion-enabled benches in 51 runs.
6	soc/debug/sim (Bash / WSL)	bash run_pulp_compare.sh	Pinned external decoder/register comparison.
7	Repository root (Bash / WSL)	bash cpu/debug/sim/run_riscv_tests.sh	Official rv32ui/rv32mi tests.
8	Repository root	python cpu/debug/sim/mutation_check.py	25 injected RTL defects.
9	cpu/debug/sim	vsim -c -do "do waves.do"	Run 8 checked benches and export VCD traces.
10	Repository root	python cpu/docs/tools/waveforms.py	SVG/PDF/PNG figures and source-window manifest.

3.4.3 Setup and configuration

Tool / setup	Requirement
ModelSim	vsim/vlog on PATH; run from the named simulation directory so file lists resolve.
Verilator	Verilator 5.050 used for the recorded run; C++ build tools, Bash and Python 3 are required.
PULP	Git and network only for the initial checkout; explicit revision pins recorded in the runner.
RISC-V GCC	A bare-metal riscv*-elf toolchain; the riscv-tests revision is pinned in the runner.
Wave rendering	Python with matplotlib; raw VCD files produced by waves.do.
Configuration	Latency/stall overrides are read back after elaboration; seeds are fixed for repeatability.

4 Annex

4.1 Verification plan

PASS indicates the recorded regression completed with its expected results and zero checker errors. Each row identifies the test that exercises the listed behavior.

ID	Title	Description	Status	Test / checker
CPU-01	CPU integration	Arithmetic, logic, load/store, trap/IRQ/WFI programs; memory/register results and protocol monitors.	PASS	rv32i_tb_cpu_axi
CPU-02	ISA reference	Independently generated reference traces: a directed program of 1,287 instructions and ten seeded random programs of 47,862 in total; nominal, delayed and stalled memories.	PASS	rv32i_tb_isa
CPU-03	ALU	ALU operations, signed/unsigned boundaries and shifts.	PASS	rv32i_tb_alu
CPU-04	Predictor / RAS	Tagged lookup, direction update, collisions, RAS push/pop/overflow/disabled setting and reset.	PASS	rv32i_tb_bp
CPU-05	Exceptions / traps	All implemented synchronous causes; precedence, wrong-path errors and precise architectural state; an interrupt and an exception on one instruction; an interrupt pending when MRET re-enables interrupts.	PASS	rv32i_tb_traps
CPU-06	CSR access	CSR legality, fixed fields, write suppression, reserved modes and reset values.	PASS	rv32i_tb_csr_ro
CPU-07	Counters	Explicit write versus increment; half preservation, carry, CSRRS/CSRRC and fixed event IDs.	PASS	rv32i_tb_counters
CPU-08	Two harts	Two distinct HART_IDs on shared memory arbitration; no result corruption.	PASS	rv32i_tb_dual_core
CPU-09	Asynchronous reset	Reset between edges during fetch/load stall, nonzero GPR state, clock stopped, and resumed execution.	PASS	rv32i_tb_reset
CPU-10	Official ISA tests	Official rv32ui/rv32mi tests at a pinned revision: 53 pass at nominal timing and 40% backpressure; five that need features outside the implemented ISA must fail.	PASS	rv32i_tb_riscv_tests

4.2 Critical sequences

The load holds S2 while its request or response is pending. At the RVALID/RREADY handshake, the data is available and s2_advance permits the instruction to enter writeback.

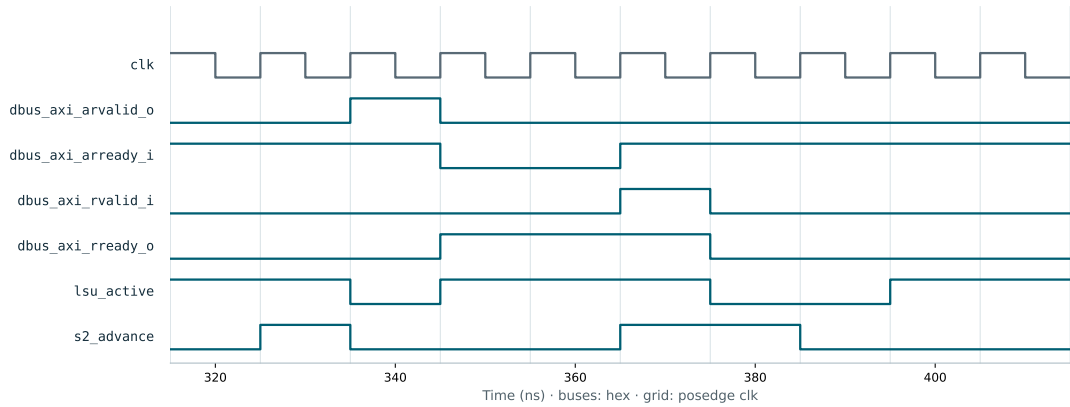


Figure 3: Load completion and pipeline advance.

The resolved branch differs from the prediction, causing redirect. IF/DX is invalidated so younger work cannot commit; an outstanding fetch response still completes its AXI transaction.

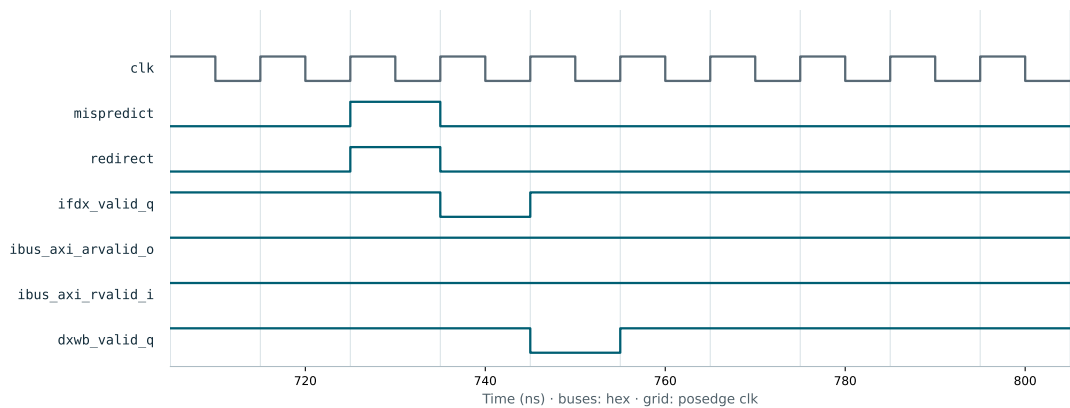


Figure 4: Branch correction after a mispredicted branch.

Scenario	Check
IRQ during load	No accepted IRQ until active LSU transaction is complete.
Wrong-path response error	Response is drained; wrong-path instruction cannot write GPR/CSR or trap.
Store channel skew	AW and W can complete in either order; one B response completes the store.

4.3 External AXI reference

The AXI4-Lite front-end is compared against a published implementation to check the interpretation of the protocol. The first table states what was compared and on what evidence; the second names every difference found and what it means.

Compared block	Shared contract	Evidence
soc_axi_lite_dec vs axi_lite_demux	Two mapped targets, MaxTrans=1; independent progress and response timing.	Compare ordered data/response records and per-target address counts.
axi_lite_slave vs axi_lite_regs	32-bit words; four RW words and one RO word; same byte strobes.	96 cases = 16 WSTRB masks × 3 AW/W orders × 2 response-stall settings.
Additional register cases	RO write, unmapped access, reset and post-reset access.	Same access-class response and no forbidden state update; each side checked against expected data.

Difference / boundary	Interpretation
Latency	Not cycle-identical; compare completed transactions and protocol compliance.
Unmapped read data	Our front-end returns zero; pinned PULP axi_lite_regs returns 0xBA5E1E55. Both return SLVERR; error-data policy is documented separately.
Address aliases	Different register-bank sizing; equality claimed only for the explicit compared address profile.
AXI4-Full	The PULP run does not establish equivalence of the DMA or soc_axi_full2lite to every upstream AXI4 module.
Converter	Local soc_tb_full2lite and soc_tb_full2lite_err check supported burst handling and errors. Local write error aggregation preserves DECERR over SLVERR; PULP is not a cycle-exact reference for that policy.
Versions	axi: 70b8e54fd460e3308e58be596ceb3566a6e3576e; common_cells: 03d98106aa19952a10360d2230def85144a0008b.

Reference: <https://github.com/pulp-platform/axi>.